

Low-Decoding-Complexity Learned Image Compression with Masked Convolutional Layer Re-parameterization

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Abstract—The high complexity of the decoding is one of the main challenges faced by Learned Image Compression (LIC). In this work, we propose the Low-Decoding-Complexity Learned Image Compression (LDCIC) framework to achieve a better trade-off between Rate-Distortion (R-D) performance and decoding complexity. Specifically, LDCIC incorporates several low-complexity transform network designs, including the Depth-wise Convolution Block, Latents at Single Low Resolution, and the Channel-Asymmetric Transform Network. In addition, we propose a Masked Convolutional Layer Re-parameterization method that compresses 2D masked kernels into equivalent 1D dense representations by identifying invariant zero regions, reducing MACs and parameters without degrading performance. Extensive experiments show that LDCIC outperforms VTM in compression performance while maintaining a decoding complexity of less than 100 kMACs/pixel.

Index Terms—Low-complexity, learned image compression, structural re-parameterization

I. INTRODUCTION

Learned Image Compression (LIC) techniques have achieved impressive rate-distortion (R-D) performance, often surpassing traditional codecs. However, many state-of-the-art LIC models still suffer from high decoding complexity. This computational burden significantly limits their practical deployment, especially on resource-constrained edge devices where power efficiency and processing speed are critical.

In this work, we propose a Low-Decoding-Complexity Learned Image Compression (LDCIC) framework that delivers competitive compression performance under low decoding complexity (less than 100 kMACs/pixel). LDCIC incorporates several low-complexity transform network designs, including the Depth-wise Convolution Block, Latents at Single Low Resolution, and the Channel-Asymmetric Transform Network. Specifically, the Depth-wise Convolution Block, adapted from DCVC-RT [1], employs depth-wise convolution, Weighted-SiLU, and Weighted-SiLU Chunk Add operations to maintain compression performance under reduced computational cost. Inspired by DCVC-RT [1], the use of Latents at Single Low Resolution enables redundancy removal at a fixed spatial scale. Compared to progressive downsampling, this approach significantly reduces peak GPU memory usage during decoding, with only minimal performance degradation. The Channel-Asymmetric Transform Network applies an analysis transform

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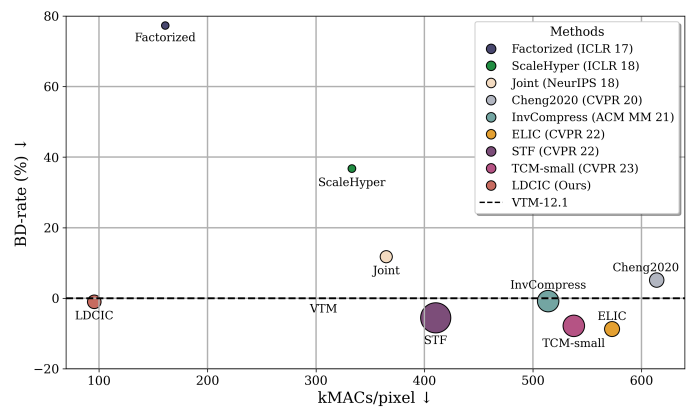


Fig. 1: Performance vs. decoding complexity on the Kodak dataset. Lower-left indicates better. Circle size denotes decoder parameters.

with more intermediate channels and a synthesis transform with fewer channels, thereby reducing the decoding complexity while maintaining model capacity. Together, these three components enable LDCIC to significantly reduce decoding complexity while preserving strong R-D performance. In addition, considering that masked convolutional layers are widely used in the entropy models of LIC and involve computational redundancy, we propose a Masked Convolutional Layer Re-parameterization technique. By removing zero-valued elements in the masked convolution kernel, the 2D convolution operation can be equivalently converted into a 1D form. This enables the convolution operation to be performed in 1D space, effectively reducing MACs and parameter count without any degradation in compression performance.

Extensive experiments on multiple benchmark datasets demonstrate that the proposed LDCIC achieves competitive compression performance under a decoding complexity of less than 100 kMACs/pixel. Notably, LDCIC achieves performance comparable to InvCompress [2] while using less than one-fifth of its decoding MACs, validating the effectiveness of our proposed design.

II. METHODOLOGY

In this section, we first introduce the architecture of the proposed Low-Decoding-Complexity Learned Image Compression (LDCIC) framework and elaborate on the core design principles behind its low-complexity decoder. We then present

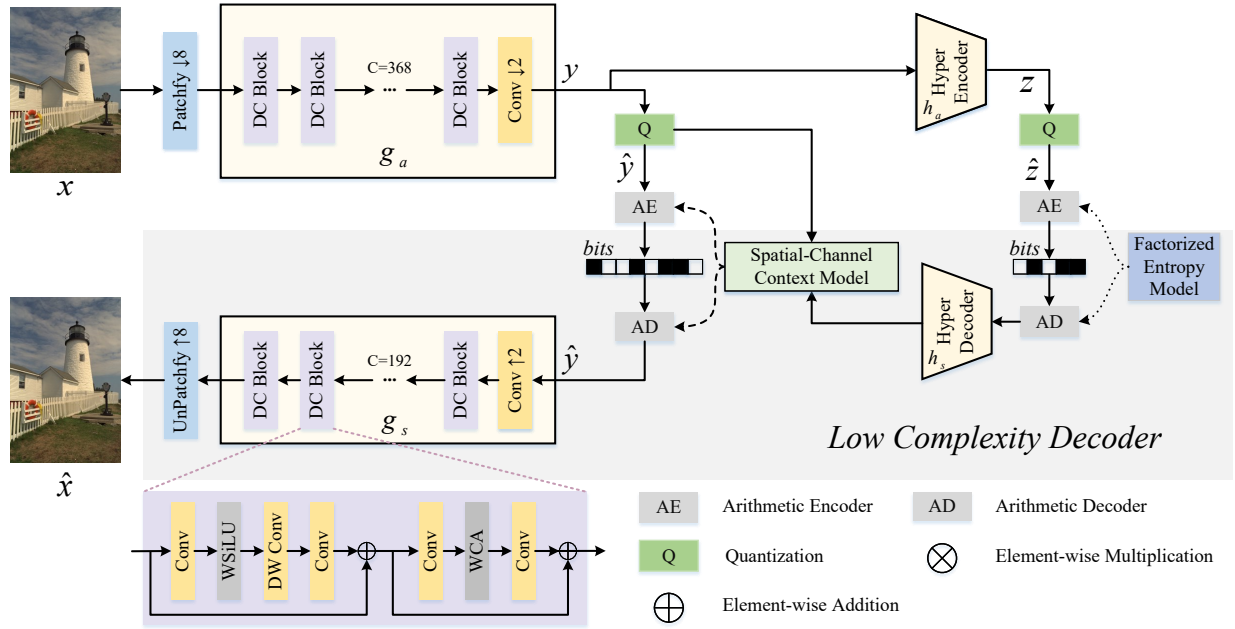


Fig. 2: The overall framework of the proposed low-decoding-complexity learned image compression.

in detail how the proposed Masked Convolutional Layer Reparameterization technique reduces the network's computational complexity via structure-equivalent transformations.

A. Network Architecture

The overall framework of the proposed method is illustrated in Figure 2. The input image x is transformed into a compact latent representation y by the main encoder g_a , and then reconstructed to the output image $\hat{x}$ by the main decoder g_s . A hyperprior encoder h_a captures the spatial dependencies in the latent representation y and generates side information z , while a hyperprior decoder h_s estimates the parameters of the entropy model from this side information. The quantized side information $\hat{z}$ is entropy coded using a factorized entropy model [3], and the quantized latent representation $\hat{y}$ is entropy coded with a space-channel context model [4].

To achieve a better trade-off between decoding complexity and rate-distortion performance, we adopt three key architectural strategies: Depth-wise Convolution Block, Latents at Single Low Resolution, and Channel-Asymmetric Transform Network Design. Furthermore, we introduce a novel technique called Masked Convolutional Layer Reparameterization, which further reduces computational complexity in a lossless manner by leveraging structurally equivalent transformations. These components will be discussed in detail in the following subsections.

B. Low-Complexity Transform Network Designs

A straightforward approach to reducing the computational complexity of the decoder is to decrease the number of intermediate feature channels in the transform networks. However, this typically results in a significant drop in rate-distortion performance. To develop a learned image compression model

that maintains high RD performance while lowering decoding complexity, we adopt three key techniques:

Depth-wise Convolution Block: Using depth-wise convolutions to reduce the computational complexity of convolutional operations has been shown to be effective in image super-resolution tasks [5]. In the field of video compression, a similar idea has recently been adopted in DCVC-RT [1], where a carefully designed Depth-wise Convolution Block (DC Block) is used as the basic block. Inspired by this, we also adopt the DC Block as the basic block to construct a low-complexity transform network. Compared to a simple depth-wise convolution, the proposed DC Block achieves significantly better performance, as shown in our ablation studies. The architecture of the DC Block is illustrated in Figure 2. In this block, WSiLU denotes a weighted SiLU activation function [6], and WCA refers to the Weighted-SiLU Chunk Add operation. These are defined as follows:

$$\begin{cases} \text{WSiLU}(x) = x \cdot \text{Sigmoid}(\alpha \cdot x), \\ \text{WCA}(x) = \sum_{k=1}^2 \text{Split}_k(\text{WSiLU}(x)), \end{cases} \quad (1)$$

where α is a weighting parameter, and Split denotes splitting the input tensor along the channel dimension, and k indexes the resulting chunks.

Latents at Single Low Resolution: Most existing LIC methods progressively downsample latents to reduce spatial dimensions, typically halving the resolution at each layer while keeping the number of channels unchanged. Recently, DCVC-RT [1], a learned video compression method, proposed learning latent representations directly at a single low resolution (i.e., 1/8 of the original image size). The authors observed that, compared to the progressive downsampling approach, this strategy significantly reduces memory I/O overhead related

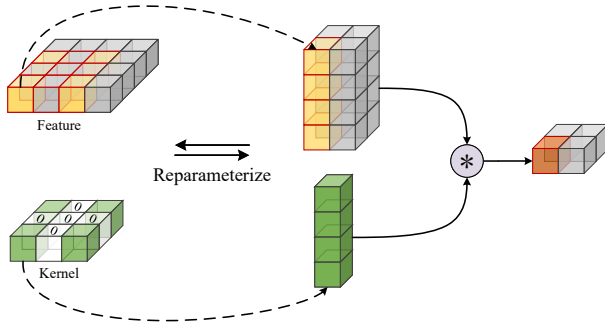


Fig. 3: The illustration of the proposed masked convolutional layer re-parameterization.

to latents and facilitates more effective latent transformation. In this paper, we further demonstrate that, compared to progressively downsampling latents, learning latents directly at a single low resolution can significantly reduce both the computational burden and peak GPU memory usage of the transform network, while incurring slightly performance degradation, as shown in our ablation studies.

Channel-Asymmetric Transform Network Design: To further reduce the computational complexity of the decoder, we adopt a channel-asymmetric transform design. Specifically, the analysis transform network g_a is configured with a larger number of intermediate feature channels, while the synthesis transform network g_s uses fewer. Concretely, we use 368 channels in the analysis transform and 192 in the synthesis transform, and both networks are composed of 13 basic blocks. This asymmetric design ensures that the analysis transform has sufficient capacity for redundancy removal, while the synthesis transform achieves a better trade-off between computational complexity and reconstruction performance.

C. Masked Convolutional Layer Re-parameterization

In the context modeling of entropy models, masked convolutional layers are widely employed to enforce spatial causality by restricting the receptive field. However, to preserve causality constraints during inference, the masked positions in the convolution kernels are filled with zeros, meaning they do not contribute to the output computation. This results in a large number of ineffective multiplications (i.e., multiplications by zero) when convolving with input features, leading to redundant computation. Due to the inherent sparsity pattern of masked convolutions, these layers often incur unnecessary memory accesses and computational inefficiencies during inference, which severely impact deployment performance. To address this problem, we propose a reparameterization technique specifically designed for masked convolutional layers.

Our approach analyzes the masked convolution kernel to identify structurally invariant zero regions and transforms the original masked convolution into an equivalent dense computation. As illustrated in Figure 3, taking a 2D 3×3 masked convolution as an example, we observe that due to the masking structure, many operations are effectively wasted during inference. By identifying the valid (i.e., non-masked) computation region, we can reparameterize the masked ker-

nel into a compact 1D representation. The original masked kernel is "compressed" into a sequence of continuous non-zero weights, and only the corresponding valid input feature elements are extracted. These are then used to perform element-wise multiplications followed by summation, yielding an output identical to that of the original masked convolution.

This reparameterization ensures functional equivalence while reducing both model parameters and computational complexity. Specifically, the computational complexity is reduced in proportion to the fraction r of non-masked weights in the original masked convolution kernel ($0 \leq r \leq 1$), as given by:

$$\text{MACS}_{\text{after}} = r \cdot \text{MACS}_{\text{before}}. \quad (2)$$

Importantly, this reparameterization is plug-and-play and requires no changes to the network architecture. It is applied only during inference, allowing the model to be trained using the original masked convolutions without any modifications. The proposed method thus maintains model functionality while substantially improving inference efficiency, making it particularly suitable for real-world deployment scenarios.

III. EXPERIMENTS

A. Experimental Setting

Training: We construct our training dataset using a total of 104,266 RGB images, sourced from the ImageNet validation set [13], DIV2K [14], Flickr2K [15], Flickr2W [16], and the first six shards of the LSDIR [17] training set. The Adam [18] optimizer is employed to minimize the R-D loss. The training process of LDCIC is divided into two stages. In the first stage, we train the model for 80 epochs using a batch size of 32 and a patch size of 256×256 , with a fixed learning rate of $1e-4$. In the second stage, we switch to a batch size of 8 and a larger patch size of 512×512 , training for 60 epochs. The learning rate is initially set to $1e-4$ and then decayed to $3e-5$, $1e-5$, $3e-6$, and $1e-6$ at the 30th, 40th, 50th, and 55th epochs, respectively. Two distortion metrics are used for optimization: MSE and MS-SSIM. When optimizing for MSE, the λ values are set to $\{0.0018, 0.0035, 0.0067, 0.0130, 0.0250, 0.0483\}$. When optimizing for MS-SSIM, the corresponding λ values are $\{2.40, 4.58, 8.73, 16.64, 31.73, 60.50\}$. We use RTX 3060 GPU to complete our experiments.

Evaluation: We evaluate the proposed method on three datasets: Kodak image set [19], Tecnick test set [20], and CLIC professional validation set [21]. Distortion is measured using both PSNR and MS-SSIM, computed on the RGB domain, while the bitrate is evaluated in terms of bits per pixel (bpp). To assess decoding complexity, we conduct a comprehensive evaluation using three metrics: Multiply-Accumulate Operations per pixel (MACs), the number of model parameters (Params), and peak GPU memory usage (GPU Mem).

Implementation Details: For our model, we set the single low resolution in the transform network to $1/8$ of the original resolution, following DCVC-RT [1]. The channel numbers in g_a and g_s are set to 368 and 192, respectively, forming a Channel-Asymmetric Transform Network. The channel dimensions of

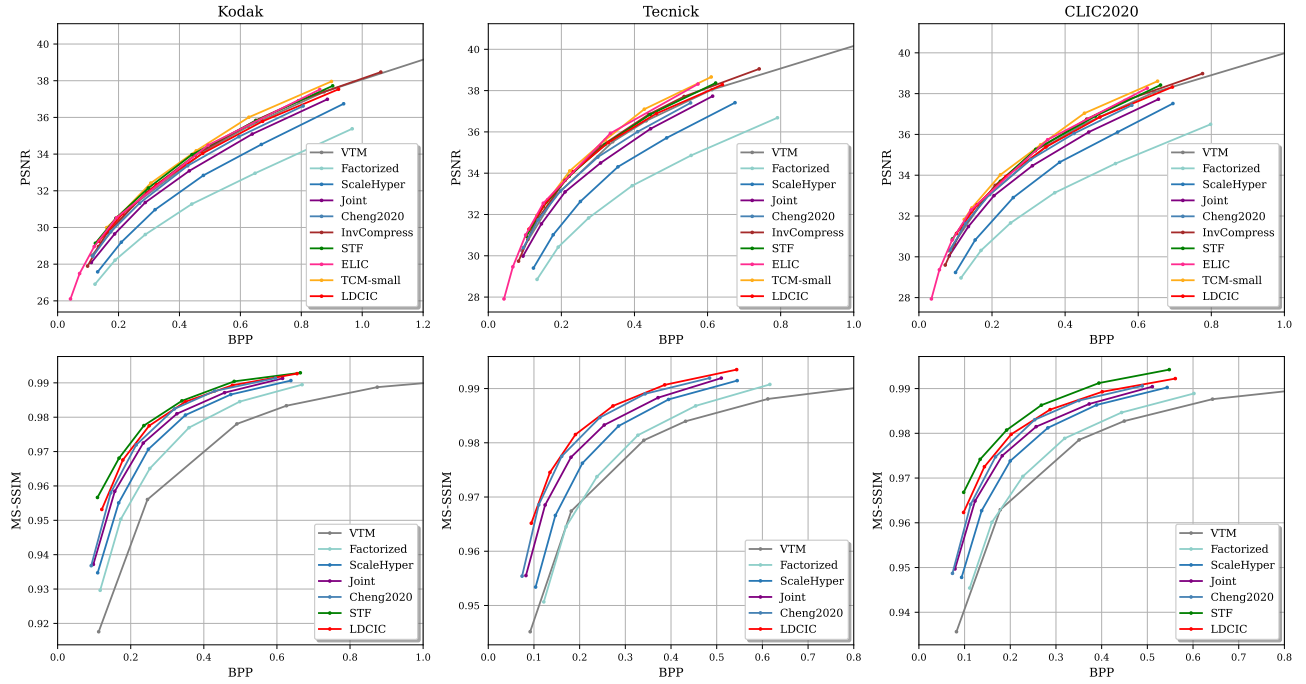


Fig. 4: Rate-Distortion curves for Kodak, Tecnick, and CLIC professional validation datasets.

TABLE I: Comparison of BD-rate (%) and Decoder Complexity. The anchor is VTM-12.1 [7] intra-coding.

Method	BD-rate (%) on Kodak		BD-rate (%) on Tecnick		BD-rate (%) on CLIC		Decoder Complexity	
	PSNR	MS-SSIM	PSNR	MS-SSIM	PSNR	MS-SSIM	MACs (K)	Params (M)
Factorized [8]	77.37	-19.18	92.42	2.27	97.29	-3.01	161.06	5.45
ScaleHyper [3]	36.81	-32.39	45.62	-18.53	44.34	-22.81	333.06	5.79
Joint [9]	11.80	-41.00	11.06	-35.27	10.67	-36.5	364.71	14.36
Cheng2020 [10]	5.18	-44.59	2.56	-42.7	2.41	-40.27	613.94	20.31
InvCompress [2]	-0.77	-	-4.49	-	-2.56	-	513.96	44.82
ELIC [4]	-8.71	-	-10.10	-	-6.82	-	572.80	21.92
STF [11]	-5.50	-53.27	-4.36	-	-3.59	-51.83	410.33	88.54
TCM-small [12]	-7.82	-	-8.73	-	-8.16	-	537.53	44.97
LDCIC	-0.98	-48.24	-5.10	-46.25	-1.78	-45.16	95.49	18.03

the latent representation y and the side information z are set to 256 and 192, respectively.

B. Experimental Results and Analysis

To evaluate the rate-distortion (R-D) performance and decoder complexity of LDCIC, we compare it against the state-of-the-art traditional codec and several representative learning-based image compression methods on three benchmark datasets. As shown in Figure 4 and Table I, in terms of PSNR-based BD-rate, LDCIC outperforms the SOTA traditional codec VTM-12.1 [7] (*encoder_intra_vtm.cfg* configuration, using YUV444 as the intermediate color space) and achieves performance comparable to InvCompress [2] across all datasets, demonstrating strong generalization capability. As shown in Figure 4, in the low bit-rate region, the R-D curve of LDCIC closely approaches that of ELIC [4]¹, while its de-

coder operates with only 16.7% of the decoding computational complexity. The low-complexity transform network designs and re-parameterized masked convolutional layer lead to a significant reduction in decoding complexity while preserving both coding efficiency.

To ensure a fair comparison of MS-SSIM optimized models, we further compare LDCIC against six representative methods whose experimental results or pre-trained MS-SSIM optimized models are available. Although LDCIC shows slightly lower performance compared to STF [11], it achieves the lowest decoding computational complexity among all evaluated methods, making it especially suitable for resource-constrained deployment scenarios where decoding efficiency is critical.

C. Ablation Studies

We conduct comprehensive experiments to validate the effectiveness of the proposed modules. The results of the ablation studies are presented in Table 2, where MCLR,

¹<https://github.com/VincentChandelier/ELiC-ReImplementation>

TABLE II: Ablation studies of the proposed modules. The anchor is LDCIC and all models are optimized for MSE.

Method	Previous Modules		Proposed Modules			Performance	Decoder Complexity		
	PDT	RBB	SLRT	DCB	MCLR	BD-rate (%) ↓	MACs (K) ↓	Params (M) ↓	GPU Mem (MB) ↓
A	✓	✓	✗	✗	✓	4.27	97.92	12.99	1821.05
B	✓	✗	✗	✓	✓	-1.16	99.31	11.79	6793.10
C	✗	✗	✓	✓	✗	0.00	99.45	18.51	1038.23
LDCIC	✗	✗	✓	✓	✓	0.00	95.49	18.03	1041.67

PDT, SLRT, RBB, and DCB refer to Masked Convolutional Layer Re-parameterization, Progressive Downsample Transform, Single Low Resolution Transform, the Residual Bottleneck Block from ELIC [4], and the Depth-wise Convolution Block from DCVC-RT [1], respectively. It is worth noting that, to align the decoding MACs with other ablation variants, Method A employs an RBB based on grouped convolution in its g_s module. Additionally, all ablation experiments are conducted on the Kodak dataset, except for GPU Mem which is measured on images with a resolution of 2560×1440 .

A comparison between Method A and Method B shows that replacing the RBB backbone in the transform network with DCB achieves a 5.43% reduction in bitrate, though at the cost of increased GPU Mem. The comparison between Method B and our proposed LDCIC demonstrates that using SLRT instead of PDT reduces GPU Mem to one-sixth, with only a 1.16% increase in bitrate. Furthermore, the comparison between Method C and LDCIC indicates that the proposed MCLR reduces decoding complexity by 3.96 kMACs/pixel and 0.48 M parameters, without compromising compression performance.

IV. CONCLUSIONS

In this paper, we present LDCIC, a low-decoding-complexity learned image compression framework. LDCIC incorporates several low-complexity transform network designs—including the Depth-wise Convolution Block, latents at single low resolution, and the channel-asymmetric transform network—to substantially reduce decoding complexity while preserving competitive compression performance. In addition, we propose a novel Masked Convolutional Layer Re-parameterization technique that achieves complexity reduction losslessly through a structurally equivalent transformation. With these carefully designed components, LDCIC achieves an effective trade-off between decoding complexity and rate-distortion performance.

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